

Non-Cartesian MRI Reconstruction

Holger Eggers^a, Melanie Kirchels^b, and Daniel Potts^b

^aPhilips Research, Hamburg, Germany

^bTechnische Universität Chemnitz, Faculty of Mathematics, Chemnitz, Germany

4.1 Introduction

As elaborated in Chapter 1, signal localization in MRI is predominantly achieved by superposing a comparatively weak magnetic field gradient onto the strong and homogeneous main magnetic field. The magnetic field gradient ideally produces a linear increase in the magnetic field strength in one direction. Since the frequency of the precession of the nuclear spins and of the signal this precession induces in the receive coils is basically proportional to the local magnetic field strength experienced by the nuclear spins, the magnetic field gradient linearly encodes the origin of the signal in the frequency. In this way, the signal is effectively measured in the spatial frequency domain, the so-called k-space.

By allowing rapid changes of the magnetic field gradient in strength and direction over time, MRI provides a remarkable flexibility in sampling k-space. The sampling position $\mathbf{k}$ at time t is simply given by the integral of the magnetic field gradient $\mathbf{G}$ over time

$$\mathbf{k}(t) = \frac{\gamma}{2\pi} \int_0^t \mathbf{G}(\tau) d\tau, \quad (4.1)$$

assuming the signal to be generated by radiofrequency excitation at $t = 0$. $\mathbf{k}$ and $\mathbf{G}$ are vectors in typically two-dimensional (2D) or three-dimensional (3D) spaces and are measured in units of $[\text{m}^{-1}]$ and $[\text{T/m}]$, respectively. The gyromagnetic ratio γ links the local magnetic field strength to the angular frequency of the precession and depends on the nucleus selected for imaging. The temporal variation of $\mathbf{G}$, commonly referred to as the gradient waveform, thus determines the k-space trajectory, along which samples are collected after a single excitation. It is only restricted by hardware constraints and more fundamentally by safety constraints, which mainly limit the velocity and acceleration in k-space. Using multiple excitations, or shots, permits segmenting the k-space trajectory by restarting in the center of k-space.

MRI was first demonstrated with projection reconstruction in 1973 [1]. This choice was presumably inspired by computed tomography (CT), with which the first clinical images had recently been obtained at that time. The data acquisition proceeded along lines through the origin and covered a polar grid in k-space, as illustrated in the bottom left plot in Fig. 4.1. According to the Fourier slice theorem, a one-dimensional (1D) inverse Fourier transform turns the samples along one of these lines into samples of a projection perpendicular to this line. This allowed employing backprojection for image reconstruction, as in CT.

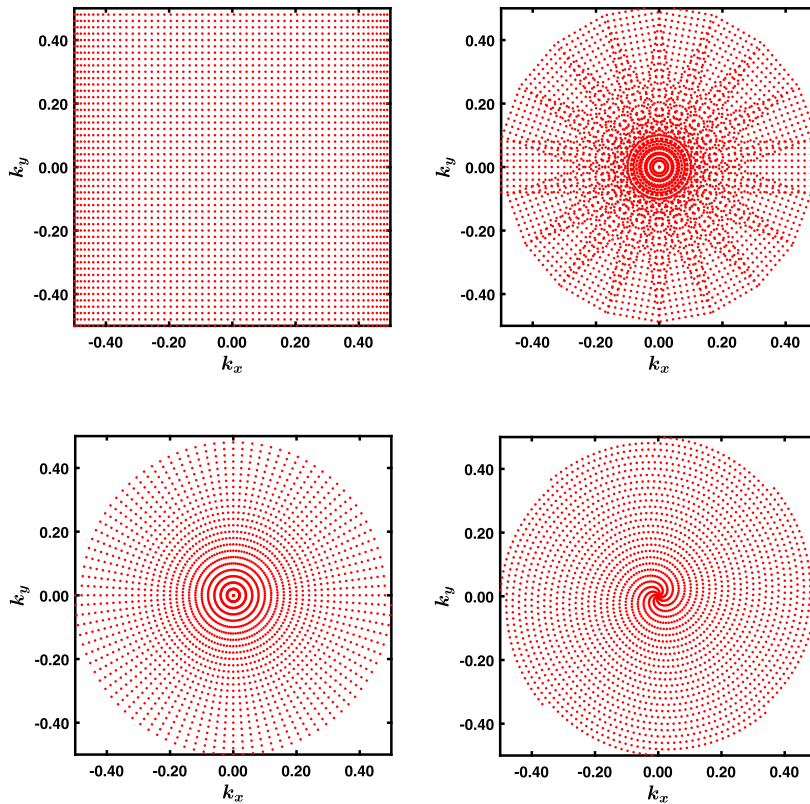


FIGURE 4.1 2D Non-Cartesian Imaging — Sampling Patterns

Nonequidistant sampling of k -space in at least one dimension is involved in 2D echo-planar (top, left), PROPELLER (top, right), radial (bottom, left), and spiral (bottom, right) imaging.

Two years later, sampling k -space on a Cartesian grid along parallel lines was proposed as a simpler alternative for MRI [2]. The application of different magnetic field gradients before and during data acquisition was introduced, establishing the distinction between phase and frequency encoding, and the use of a multidimensional inverse fast Fourier transform (FFT) for image reconstruction was suggested. As later remarked by one of the authors of this seminal work, this was driven by the conviction that projections cannot efficiently cover a 2D plane or 3D volume due to the resulting strong variation in sampling density between the center and the periphery of k -space [3]. The combination of sampling k -space on a Cartesian grid and reconstructing images based on an inverse FFT, known as Cartesian imaging, soon developed into the prevalent method for MRI. In particular in clinical practice, this still holds today.

Nevertheless, a considerable variety of methods that deliberately acquires data not on a Cartesian grid has been devised and explored over the last decades. The sampling patterns obtained in k -space

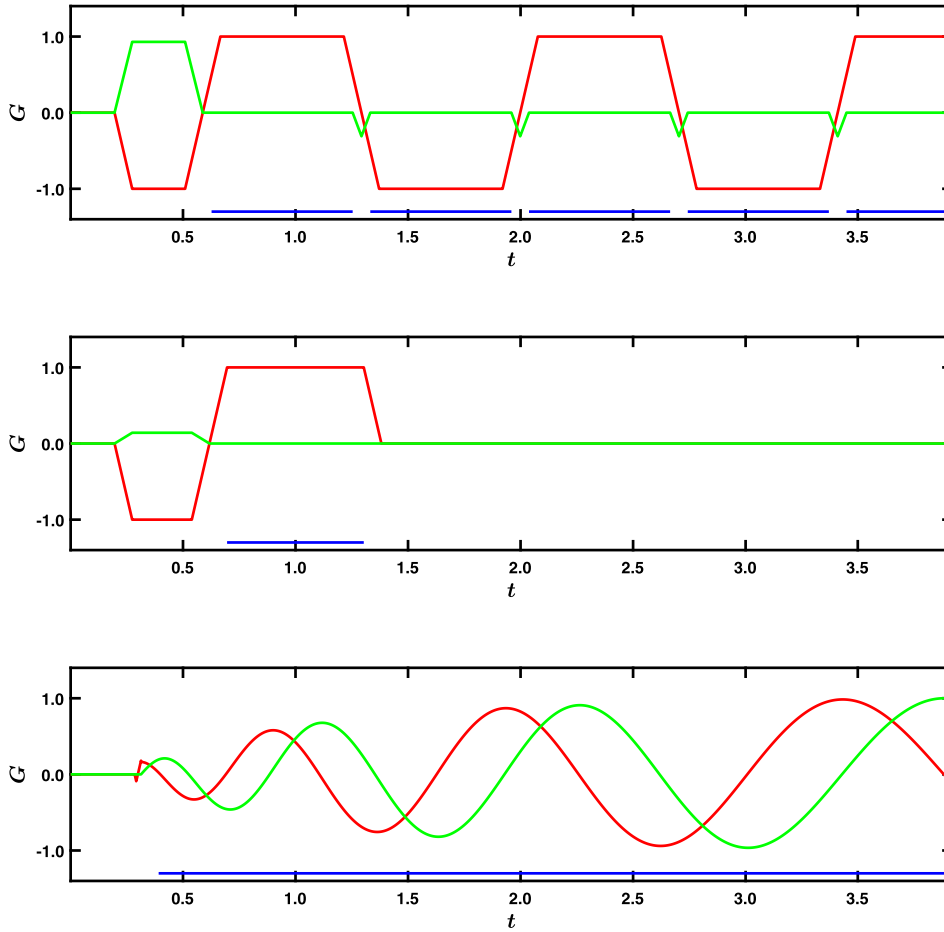


FIGURE 4.2 2D Non-Cartesian Imaging — Gradient Waveforms

2D echo-planar (top), PROPELLER or radial (middle), and spiral (bottom) imaging rely on different gradient waveforms $G(t)$ in the frequency- and phase-encoding directions (red, green (mid gray, gray in print version)) and on different acquisition windows (blue (dark gray in print version)) to generate the sampling patterns in Fig. 4.1. For the sake of simplicity, they are all assumed to be based on gradient echoes.

with four of the most popular of these methods in two dimensions, including projection reconstruction, or radial imaging, are plotted in Fig. 4.1.

Echo-planar imaging (EPI) is often not even considered as a non-Cartesian imaging method [4]. It employs a gradient waveform consisting of a train of trapezoids with alternating polarity in the frequency-encoding direction. The beginning of such a train is shown in the top plot in Fig. 4.2, where G and t are both normalized. In its purest form, the whole k -space is traversed after a single excitation along parallel lines in opposite directions. To increase speed and efficiency further, the first samples

along each line are already collected while the magnetic field gradient is still ramped up, and the last samples are still collected while the magnetic field gradient is already ramped down again. This so-called ramp sampling leads to an increasing sampling density towards the edges of k-space in the frequency-encoding direction, while a constant sampling density is preserved everywhere else. The data acquisition is only interrupted for switching between the parallel lines by application of a magnetic field gradient in the phase-encoding direction and is not started before a corner of k-space has been reached by means of the respective first trapezoid in each direction. EPI is in widespread use today, especially for diffusion-weighted imaging (DWI) and functional imaging (fMRI).

Periodically rotated overlapping parallel lines with enhanced reconstruction (PROPELLER) imaging is also a hybrid Cartesian and non-Cartesian imaging method [5]. The data acquisition is split up into blades, with each of which a rectangular k-space area is covered. Each blade supports the reconstruction of an intermediate image with anisotropic spatial resolution based on an inverse FFT. These intermediate images enable detection and correction of motion between the acquisition of the individual blades, which are rotated about the k-space origin with respect to each other. However, the reconstruction of the final image has to cope with the highly nonequidistant sampling of k-space by all blades together. The gradient waveforms are identical to those in Cartesian imaging, except that the frequency- and phase-encoding directions are rotated with the blades. PROPELLER imaging is of particular relevance in applications prone to motion, including head, neck, spine, and shoulder imaging.

Radial imaging is primarily employed in areas in which the strong variation in sampling density between the center and the periphery of k-space actually promises advantages. Above all, the oversampling in the center leads to insensitivity to motion by averaging out inconsistencies in the data to a large extent. Moreover, residual artifacts are spread in two dimensions and appear as streaks, which are usually more benign than the coherent or incoherent ghosts arising along one dimension in Cartesian imaging without variation in sampling density. Additionally, the repeated coverage of the center enables image reconstruction at various spatial and temporal resolutions in dynamic imaging, as well as retrospective self-gating in abdominal and cardiac imaging during free breathing without interspersed navigators or dedicated sensors. Besides, radial imaging allows dispensing with phase encoding by starting the data acquisition in the center, permitting very short echo times to capture rapidly decaying signal, as required for lung and bone imaging. Compared to PROPELLER imaging, the gradient waveform in the phase-encoding direction is eliminated, and the gradient waveform in the frequency-encoding direction is rotated with the projections. The first trapezoid in the frequency-encoding direction is kept to obtain full projections or is removed to obtain half projections, as demonstrated in the later examples.

Spiral imaging first and foremost provides a fast and flexible coverage of k-space [6]. It supports single-shot and multishot imaging, uniform sampling density in all but the very center of k-space and variable sampling density across k-space, as well as traversing k-space from the center outward and from the periphery inward. Unlike in EPI, the magnetic field gradient continuously changes during the data acquisition, and dead times for switching its polarity are eliminated. Furthermore, motion and aliasing artifacts are rather incoherent, and especially flow artifacts are less pronounced. Spiral imaging is mainly of interest in a wide range of applications demanding high speed and efficiency, but is very susceptible to an imperfect main magnetic field or magnetic field gradient.

All of these 2D methods can be extended to three dimensions, either by the so-called stacking, the addition of an equidistant sampling along the third dimension, or by a suitable generalization.

Due to the diversity of sampling patterns in MRI, general algorithms are favored for the reconstruction in non-Cartesian imaging. In view of the limited signal-to-noise ratio (SNR) attainable with MRI,

their accuracy only has to be moderate, in particular in comparison with CT. An essential part of such algorithms, a nonequispaced fast Fourier transform (NFFT), is described in the next section. Based on it, ubiquitous gridding is introduced in the following section.

4.2 NFFT

The signal s measured in k-space is, as in Chapter 2, modeled by

$$s(k) = \int_{-N/2}^{N/2} x(r) e^{-i2\pi kr} dr, \quad (4.2)$$

where x denotes the transverse magnetization of an object of maximum extent $[-\frac{N}{2}, \frac{N}{2}]$, from now on simply referred to as the object and the field of view (FOV).

An FFT of length N efficiently converts N equidistant samples of x into N equidistant samples of $\tilde{s}$ defined by

$$\tilde{s}_m = \sum_{n=1}^N x_n e^{-i2\pi k_m r_n}. \quad (4.3)$$

Here, r and k are unitless and normalized to the range $[-\frac{N}{2}, \frac{N}{2}]$ and $[-\frac{1}{2}, \frac{1}{2}]$, respectively. Assuming N to be an even positive integer, the sampling positions in space are then located at $r_n = -\frac{N}{2}, -\frac{N}{2} + 1, \dots, \frac{N}{2} - 1$. $\tilde{s}$ is linked to s by

$$\tilde{s}(k) = \sum_{p=-\infty}^{\infty} s(k + p) \quad (4.4)$$

due to the limited coverage of k-space by the acquisition and the spatial discretization of the object in the reconstruction. Thus, the $\tilde{s}_m$ in Eq. (4.3) are not samples of s , but of s turned into a periodic signal $\tilde{s}$ with a period of 1. To reduce the computational complexity from $\mathcal{O}(N^2)$ of a direct evaluation of Eq. (4.3) to $\mathcal{O}(N \log N)$, an FFT exploits symmetries in the exponential factors, which are lost if the r_n or k_m are not equidistant anymore.

An NFFT generalizes an FFT to nonequidistant samples of x , $\tilde{s}$, or both [7,8]. It relies on an FFT for the actual transformation between the spatial and the spatial frequency domain and, therefore, has to convert nonequidistant to equidistant samples intermediately. This involves an approximation, the accuracy of which can be traded off for computational complexity. In the following, the case of equidistant samples of x and nonequidistant samples of $\tilde{s}$ is considered. It is of relevance for an efficient evaluation of Eq. (4.3), the forward model, which requires in non-Cartesian imaging the calculation of M samples of $\tilde{s}$ from N samples of x . The other two cases are discussed at the end of this chapter.

To convert equidistant into nonequidistant samples in k-space, a convolution with a window function c , turned into a periodic function $\tilde{c}$ with a period of 1 by

$$\tilde{c}(k) = \sum_{p=-\infty}^{\infty} c(k + p), \quad (4.5)$$

is performed. $\tilde{c}$ is assumed to have a uniformly convergent Fourier series, which allows writing it as

$$\tilde{c}(k) = \sum_{r=-\infty}^{\infty} \tilde{c}_r e^{i2\pi kr} \quad (4.6)$$

with the Fourier coefficients

$$\tilde{c}_r = \int_{-1/2}^{1/2} \tilde{c}(k) e^{-i2\pi kr} dk. \quad (4.7)$$

Substituting k by $k - k'$ results in

$$\tilde{c}_r = \int_{-1/2}^{1/2} \tilde{c}(k - k') e^{-i2\pi (k - k')r} dk', \quad (4.8)$$

which is approximated by

$$\tilde{c}_r \approx \frac{1}{fN} \sum_{n'=-fN/2}^{fN/2-1} \tilde{c}\left(k - \frac{n'}{fN}\right) e^{-i2\pi \left(k - \frac{n'}{fN}\right)r}. \quad (4.9)$$

The oversampling factor $f \geq 1$ is introduced to increase accuracy, where fN is also assumed to be an even positive integer [9,10]. If the periodic function $\tilde{c}$ is bandlimited with a maximum frequency of $\frac{fN}{2}$, Eq. (4.9) holds exactly. Otherwise, aliasing occurs, and the estimation of the $\tilde{c}_r$ is confounded by higher frequencies. More specifically, the frequency bands $[fNp - \frac{N}{2}, fNp + \frac{N}{2})$, for any integer $p \neq 0$, fold back onto the baseband $[-\frac{N}{2}, \frac{N}{2})$. Rearranging Eq. (4.9) leads to

$$e^{i2\pi kr} \approx \frac{1}{fN\tilde{c}_r} \sum_{n'=-fN/2}^{fN/2-1} \tilde{c}\left(k - \frac{n'}{fN}\right) e^{i2\pi \frac{n'}{fN}r}, \quad (4.10)$$

provided that $\tilde{c}_r \neq 0$, and substituting r by $-r$ to

$$e^{-i2\pi kr} \approx \frac{1}{fN\tilde{c}_{-r}} \sum_{n'=-fN/2}^{fN/2-1} \tilde{c}\left(k - \frac{n'}{fN}\right) e^{-i2\pi \frac{n'}{fN}r}, \quad (4.11)$$

which is inserted into Eq. (4.3) to obtain

$$\tilde{s}_m \approx \sum_{n'=-fN/2}^{fN/2-1} \tilde{c}\left(k_m - \frac{n'}{fN}\right) \sum_{n=1}^N \frac{x_n}{fN\tilde{c}_{-r_n}} e^{-i2\pi \frac{n'}{fN}r_n}. \quad (4.12)$$

The inner sum amounts to an FFT of length fN , transforming N equidistant samples of x in space, after weighting and zero padding to fN equidistant samples in space, into fN equidistant samples in k-space. The outer sum converts the latter by a convolution into M nonequidistant samples in k-space.

Choosing

$$\tilde{c}_r = \begin{cases} \frac{1}{fN}, & -\frac{fN}{2} < r \leq \frac{fN}{2} \\ 0, & \text{otherwise} \end{cases} \quad (4.13)$$

allows dispensing with the weighting, usually referred to as deapodization, and results with Eq. (4.6) in

$$\tilde{c}(k) = \frac{1}{fN} \sum_{r=-fN/2+1}^{fN/2} e^{i2\pi kr}. \quad (4.14)$$

Here, the shift in the range of r arises from substituting r by $-r$ in Eq. (4.11). $\tilde{c}(k)$ is a complex trigonometric polynomial of degree $\frac{fN}{2}$ in this case, for which Eq. (4.9) holds exactly. It is written as

$$\tilde{c}(k) = \begin{cases} \frac{\sin(\pi k f N)}{fN \sin(\pi k)} e^{i\pi k}, & k \neq 0 \\ 1, & \text{otherwise} \end{cases} \quad (4.15)$$

and approximated by

$$\tilde{c}(k) \approx \text{sinc}(\pi k f N) \quad (4.16)$$

for small πk using

$$\text{sinc}(x) = \begin{cases} \frac{\sin(x)}{x}, & x \neq 0 \\ 1, & \text{otherwise} \end{cases}. \quad (4.17)$$

According to the sampling theorem, s can be recovered from equidistant samples of s by

$$s(k) = \sum_{n=-\infty}^{\infty} s\left(\frac{n}{fN}\right) \text{sinc}\left(\pi\left(k - \frac{n}{fN}\right)fN\right). \quad (4.18)$$

Consequently, the sinc function of infinite extent is the ideal window function in the hypothetical case of unlimited coverage of k -space by the acquisition. In the practicable case of limited coverage of k -space, however, it is, according to Eq. (4.15), better replaced by the sinc function turned into a periodic function, multiplied by the phasor $e^{i\pi k}$ arising from the asymmetry in the r_n for even fN , which is known as the Dirichlet kernel.

To reduce the computational complexity of the convolution, the window function is preferably real and limited in its extent, either by design or by truncation, to $[-\frac{K}{2fN}, \frac{K}{2fN}]$ with a positive integer K , the kernel size. A truncation leads to

$$c'(k) = \begin{cases} c(k), & |k| \leq \frac{K}{2fN} \\ 0, & \text{otherwise} \end{cases} \quad (4.19)$$

and a corresponding periodic function $\tilde{c}'$ with a period of 1, and it changes Eq. (4.12) to

$$\tilde{s}_m \approx \sum_{n'=-fN/2}^{fN/2-1} \tilde{c}'\left(k_m - \frac{n'}{fN}\right) \sum_{n=1}^N \frac{x_n}{fN \tilde{c}_{-r_n}} e^{-i2\pi \frac{n'}{fN} r_n}. \quad (4.20)$$

Prolate spheroidal wave functions maximize the relative energy of a function of given extent in one domain in an interval of given extent in the other domain [11]. The Kaiser–Bessel window is, in the continuous case, a simple approximation to them [12]. Thus, it is particularly well-suited to minimize aliasing errors in Eq. (4.12) and truncation errors in Eq. (4.20) [10,13]. If aliasing errors are to be minimized, the deapodization is defined by the Kaiser–Bessel window

$$\tilde{c}_r = \begin{cases} \frac{1}{fN} I_0 \left(b \sqrt{1 - \left(\frac{\pi K}{fNb} r \right)^2} \right), & |r| \leq \frac{fNb}{\pi K} \\ 0, & \text{otherwise} \end{cases}, \quad (4.21)$$

where I_0 is the zeroth order modified Bessel function of the first kind. By setting the shape parameter b to

$$b = \pi K \left(1 - \frac{1}{2f} \right), \quad (4.22)$$

the Fourier coefficients drop to zero for $|r| > fN - \frac{N}{2}$ [14]. The window function, also referred to as the convolution kernel, is then given by

$$c(k) = \begin{cases} \frac{2b}{\pi K} \text{sinhc} \left(b \sqrt{1 - \left(\frac{2fN}{K} k \right)^2} \right), & |k| \leq \frac{K}{2fN} \\ \frac{2b}{\pi K} \text{sinc} \left(b \sqrt{\left(\frac{2fN}{K} k \right)^2 - 1} \right), & \text{otherwise} \end{cases} \quad (4.23)$$

using

$$\text{sinhc}(x) = \begin{cases} \frac{\sinh(x)}{x}, & x \neq 0 \\ 1, & \text{otherwise} \end{cases}. \quad (4.24)$$

If truncation errors are to be minimized instead, the convolution kernel is defined by the Kaiser–Bessel window

$$c(k) = \begin{cases} \frac{1}{K} I_0 \left(b \sqrt{1 - \left(\frac{2fN}{K} k \right)^2} \right), & |k| \leq \frac{K}{2fN} \\ 0, & \text{otherwise} \end{cases}, \quad (4.25)$$

and the deapodization is given by

$$\tilde{c}_r = \begin{cases} \frac{1}{fN} \text{sinhc} \left(b \sqrt{1 - \left(\frac{\pi K}{fNb} r \right)^2} \right), & |r| \leq \frac{fNb}{\pi K} \\ \frac{1}{fN} \text{sinc} \left(b \sqrt{\left(\frac{\pi K}{fNb} r \right)^2 - 1} \right), & \text{otherwise} \end{cases}. \quad (4.26)$$

Both choices lead to similar convolution kernels and deapodizations, as illustrated in Fig. 4.3 and Fig. 4.4 with $f = 2.0$, $K = 5$, $N = 800$, and a normalization of the values of the functions to the range

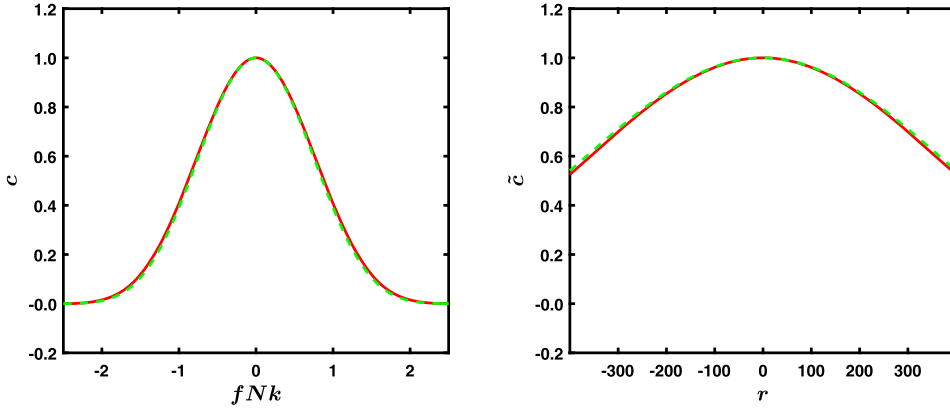


FIGURE 4.3 Convolution Kernels and Deapodizations

The Fourier transform of a Kaiser–Bessel window is employed as a convolution kernel in the conversion of equidistant into nonequidistant samples in k -space (solid red (mid gray in print version), left). To compensate for the inherent low-pass filtering, a deapodization is performed in space by dividing by the Kaiser–Bessel window (solid red (mid gray in print version), right). Alternatively, a Kaiser–Bessel window serves as convolution kernel (dashed green (gray in print version), left) and dividing by the inverse Fourier transform of the Kaiser–Bessel window as deapodization (dashed green (gray in print version), right).

[0, 1]. It is worth noting that the second case in Eq. (4.23) and Eq. (4.26) is only provided for completeness. Due to the truncation of the convolution kernel and the limited extent of the object, only the first case is of relevance in practice. Moreover, by subtracting an offset of $\frac{1}{fN}I_0(b)$ and $\frac{1}{K}I_0(b)$ from the first case of Eq. (4.21) and Eq. (4.25), the Kaiser–Bessel window smoothly drops to zero already at $|r| = \frac{fNb}{\pi K}$ and $|k| = \frac{K}{2fN}$, which enables eliminating aliasing errors and preventing the convolution kernel from extending over $K + 1$ equidistant samples in k -space in the worst case, respectively.

In general, the accuracy and the computational complexity of an NFFT increase with the oversampling factor and the kernel size. Error bounds have been established for various window functions to guide the selection of minimal values for f and K given a target accuracy [8]. Since the running time of an FFT does not monotonically increase with the length of the FFT, the choice of f is preferably made in view of the running time of an FFT of length fN [15]. The complexity of the window function is less of a concern because it is typically calculated only once and stored in a look-up table [16].

Eq. (4.20) can be rewritten as

$$\tilde{\mathbf{s}} \approx \mathbf{C}' \mathbf{F} \mathbf{D} \mathbf{x} \quad (4.27)$$

with a diagonal $N \times N$ deapodization matrix $\mathbf{D}$

$$[\mathbf{D}]_{n,n} = \frac{1}{fN\tilde{c}_{-r_n}}, \quad (4.28)$$

an $fN \times N$ Fourier transform matrix $\mathbf{F}$

$$[\mathbf{F}]_{n',n} = e^{-i2\pi \left(\frac{n'-1}{fN} - \frac{1}{2} \right) r_n} \quad (4.29)$$

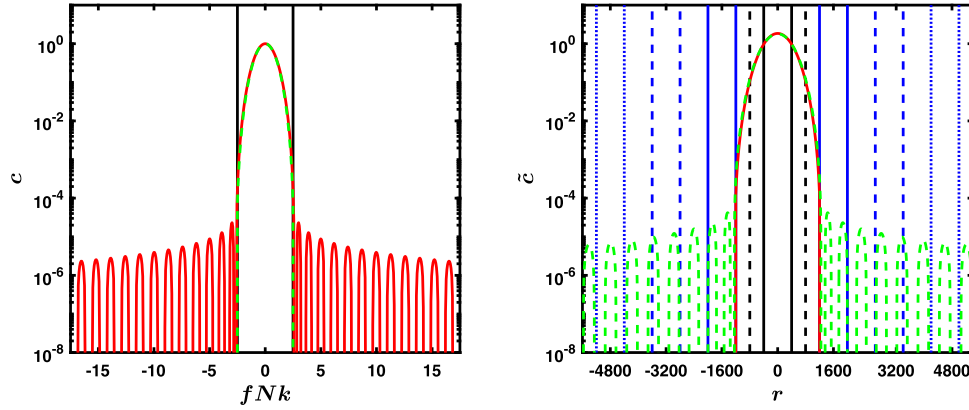


FIGURE 4.4 Aliasing and Truncation Errors

The convolution kernels from Fig. 4.3 and their inverse Fourier transforms are displayed with a logarithmic scale over a larger range of k and r , respectively. While a Kaiser–Bessel window is a function with compact support, its Fourier transform and inverse Fourier transform are not. Consequently, truncation errors occur in k -space using a Kaiser–Bessel window for deapodization in space (solid red (mid gray in print version)), whereas aliasing errors occur in space using it as convolutional kernel in k -space (dashed green (gray in print version)). The prescribed kernel size and FOV (solid black, left and right), the FOV extended by oversampling (dashed black, right), and the first three frequency bands folding back onto the prescribed FOV (solid, dashed, and dotted blue (dark gray in print version), right) are indicated.

and a sparse $M \times fN$ convolution matrix $\mathbf{C}'$

$$[\mathbf{C}']_{m,n'} = \tilde{c}' \left(k_m - \frac{n'-1}{fN} + \frac{1}{2} \right). \quad (4.30)$$

Here, the entries in the columns and rows of the matrices are numbered starting at 1, which complicates the expressions in the parentheses in Eq. (4.29) and Eq. (4.30) compared to Eq. (4.20). Eq. (4.27) is an approximation of

$$\tilde{\mathbf{s}} = \mathbf{E} \mathbf{x} \quad (4.31)$$

with an $M \times N$ encoding matrix $\mathbf{E}$

$$[\mathbf{E}]_{m,n} = e^{-i2\pi k_m r_n}. \quad (4.32)$$

Throughout this chapter, M and N are assumed to be of the same order of magnitude, ignoring Fourier interpolation for the moment. Undersampling is the subject of subsequent chapters.

Extending an NFFT to multiple dimensions is straightforward by using a tensor product approach. While separable convolution kernels, defined by the product of 1D convolution kernels for each dimension, are convenient and prevailing in practice, nonseparable convolution kernels, such as circularly and spherically symmetric ones, promise to enhance accuracy given a target computational complexity [17]. However, a theoretical comparison of them still seems to be missing.

4.3 Gridding

The object x is ideally recovered from the signal s by

$$x(r) = \int_{-\infty}^{\infty} s(k) e^{i2\pi kr} dk. \quad (4.33)$$

Due to the limited coverage of k -space by the acquisition, s is only known for k in $[-\frac{1}{2}, \frac{1}{2})$ and, without prior knowledge, disregarded elsewhere, which leads to the approximation

$$\hat{x}(r) = \int_{-1/2}^{1/2} s(k) e^{i2\pi kr} dk. \quad (4.34)$$

In addition, s is sampled with finite resolution, limited by hardware constraints in the frequency-encoding direction and more fundamentally by time constraints in the phase-encoding direction, resulting in the further approximation

$$\hat{x}_n = \frac{1}{M} \sum_{m=1}^M s_m e^{i2\pi k_m r_n} \quad (4.35)$$

or

$$\hat{\mathbf{x}} = \frac{1}{M} \mathbf{E}^H \mathbf{s}. \quad (4.36)$$

An inverse FFT of length M efficiently converts M equidistant samples of s into M equidistant samples of $\hat{x}$. If the object is to be reconstructed with higher spatial resolution, that is to be enlarged by Fourier interpolation, the number of equidistant samples of $\hat{x}$, and with it the length of the inverse FFT, is increased here to $N > M$, whereas the spacing of the equidistant samples of s is decreased to $\frac{1}{N}$. In analogy with Eq. (4.14) and Eq. (4.15), the point spread function (PSF), which links x to $\hat{x}$, is then given by

$$\begin{aligned} \text{PSF}(r) &= \frac{1}{M} \sum_{m=-M/2}^{M/2-1} e^{i2\pi \frac{m}{N} r} \\ &= \frac{\sin(\pi \frac{r}{N} M)}{M \sin(\pi \frac{r}{N})} e^{-i\pi \frac{r}{N}}, \end{aligned} \quad (4.37)$$

where the scaling is chosen such that $\text{PSF}(0) = 1$. Thus, the equidistant samples of $\hat{x}$ are approximately related by a sinc interpolation. Only in the case of $N = M$, which is exceptional, because Fourier interpolation is almost always applied in clinical routine, they are uncorrelated.

An inverse FFT is both the adjoint and the inverse of the corresponding FFT. For an NFFT, this does not hold anymore. Gridding was developed in radioastronomy from simpler interpolation methods and adopted in medical imaging before NFFTs were proposed and thoroughly analyzed in mathematics [18,9]. It aims at reconstructing N equidistant samples of x from M nonequidistant samples of s . To provide an approximate explicit inverse NFFT, gridding relies on an adjoint NFFT and introduces

an additional weighting of the nonequidistant samples of s , the sampling density compensation. It is described by

$$\hat{\mathbf{x}} \approx \mathbf{D}^H \mathbf{F}^H \mathbf{C}'^H \mathbf{W} \mathbf{s} \quad (4.38)$$

with an $M \times M$ diagonal matrix $\mathbf{W}$ of a vector of weights $\mathbf{w}$. Accordingly, it involves, after the sampling density compensation, a convolution with a window function, an inverse FFT, and a deapodization. Compared to the NFFT derived in the previous section, these three steps are performed in reverse order. Moreover, the convolution is evaluated on an oversampled Cartesian grid, the FFT is replaced by an inverse FFT, and the extended FOV is cropped by the deapodization.

Various approaches to deriving a suitable sampling density compensation have been proposed. One class of approaches considers each s_m as representative for a certain neighborhood in k -space and turns Eq. (4.35) into the Riemann sum

$$\hat{x}_n = \sum_{m=1}^M w_m s_m e^{i2\pi k_m r_n}. \quad (4.39)$$

Here, the sum of the positive weights that correspond to lengths in one dimension is normalized to a value of 1. The calculation of the weights is based on either a continuous or a discrete model.

For simple non-Cartesian sampling patterns, the weights can analytically be obtained from the Jacobian determinant of a differentiable coordinate transformation. In the case of 2D radial imaging, the coordinate transformation from a polar grid to a Cartesian grid is defined by

$$\begin{aligned} k_x &= k_r \cos(2\pi k_\phi) \\ k_y &= k_r \sin(2\pi k_\phi) \end{aligned} \quad (4.40)$$

with the radial direction k_r and the azimuthal direction k_ϕ normalized to the range $[0, \frac{1}{2}]$ and $[-\frac{1}{2}, \frac{1}{2})$, respectively. The determinant of the Jacobian matrix

$$J_R = \begin{bmatrix} \frac{\partial k_x}{\partial k_r} & \frac{\partial k_x}{\partial k_\phi} \\ \frac{\partial k_y}{\partial k_r} & \frac{\partial k_y}{\partial k_\phi} \end{bmatrix} \quad (4.41)$$

is then equal to $2\pi k_r$. Extending the integration in Eq. (4.34) to a circular area R of radius $\frac{1}{2}$ in two dimensions results in

$$\hat{x}(r_x, r_y) = \iint_R s(k_x, k_y) e^{i2\pi(k_x r_x + k_y r_y)} dk_x dk_y \quad (4.42)$$

and a change in variables in

$$\hat{x}(r_x, r_y) = 2\pi \int_0^{1/2} \int_{-1/2}^{1/2} s(k_r, k_\phi) k_r e^{i2\pi k_r (\cos(2\pi k_\phi) r_x + \sin(2\pi k_\phi) r_y)} dk_\phi dk_r. \quad (4.43)$$

Accordingly, $s(k_r, k_\phi)$ is weighted by $2\pi k_r$, which corresponds to the ideal ramp filter used in back-projection.

In the case of 2D spiral imaging, a coordinate transformation to a polar grid exists for some common k-space trajectories, which can readily be extended to a coordinate transformation to a Cartesian grid [19]. For a segmented, or interleaved, variable angular speed k-space trajectory, as illustrated in the bottom right plot in Fig. 4.1, it is defined by

$$\begin{aligned} k_r &= \frac{k_s}{2\sqrt{\alpha + (1-\alpha)k_s}} \\ k_\phi &= \frac{N}{M_I} \cdot \frac{k_s}{2\sqrt{\alpha + (1-\alpha)k_s}} + k_i, \end{aligned} \quad (4.44)$$

where k_s parameterizes the sample, or time, along an interleaf, normalized to the range $[0, 1]$, and k_i parameterizes the interleaf, or its rotation about the k-space origin, normalized to the range $[-\frac{1}{2}, \frac{1}{2})$. Here, both k_s and k_i are considered as continuous. The range of k_ϕ is extended to $[-\frac{1}{2}, \frac{1}{2} + \frac{N}{2M_I})$, and M is subdivided into M_I interleaves with M_S samples each. Since

$$k_r = \frac{M_I}{N} (k_\phi - k_i), \quad (4.45)$$

this is an Archimedean spiral with a parameter α in the range $[0, 1]$. For $\alpha = 0$ and $\alpha = 1$, a constant linear velocity spiral and a constant angular velocity spiral is obtained, respectively. The determinant of the Jacobian matrix

$$J_S = \begin{bmatrix} \frac{\partial k_r}{\partial k_s} & \frac{\partial k_r}{\partial k_i} \\ \frac{\partial k_\phi}{\partial k_s} & \frac{\partial k_\phi}{\partial k_i} \end{bmatrix} \quad (4.46)$$

is then given by

$$|J_S| = \frac{2\alpha + (1-\alpha)k_s}{4(\alpha + (1-\alpha)k_s)^{\frac{3}{2}}} \quad (4.47)$$

and the determinant of the Jacobian matrix of the composite coordinate transformation to a Cartesian grid by

$$|J_S J_R| = \frac{\pi}{4} \cdot \frac{2\alpha k_s + (1-\alpha)k_s^2}{(\alpha + (1-\alpha)k_s)^2}. \quad (4.48)$$

For more complex non-Cartesian sampling patterns, the weights can geometrically be obtained from the discrete k-space sampling positions by constructing a Voronoi diagram and calculating the area of each Voronoi cell [20].

Another class of approaches solves a minimization problem. For example, replacing $\mathbf{s}$ in Eq. (4.38) by $\tilde{\mathbf{s}}$ from Eq. (4.27) leads to

$$\hat{\mathbf{x}} \approx \mathbf{D}^H \mathbf{F}^H \mathbf{C}'^H \mathbf{W} \mathbf{C}' \mathbf{F} \mathbf{D} \mathbf{x}. \quad (4.49)$$

In analogy with Eq. (4.13) and Eq. (4.14), turning the sparse convolution matrix $\mathbf{C}'$ into a dense convolution matrix $\mathbf{C}$ allows eliminating the deapodization by setting $\mathbf{D} = \mathbf{I}$. Choosing additionally $f = 1.0$ then simplifies Eq. (4.49) to

$$\mathbf{F} \hat{\mathbf{x}} \approx \mathbf{N} \mathbf{C}^H \mathbf{W} \mathbf{C} \mathbf{F} \mathbf{x} \quad (4.50)$$

since $\mathbf{F}\mathbf{F}^H = N\mathbf{I}$. Ideally, applying the forward model to an image and then gridding to the simulated signal recovers the image. Independent of $\mathbf{x}$, this would be ensured by the condition

$$N\mathbf{C}^H\mathbf{W}\mathbf{C} = \mathbf{I}. \quad (4.51)$$

However, this matrix equation is normally overdetermined. It can be rewritten as a linear system of N^2 equations and multiplied on the left with the system matrix, or the Frobenius norm of the matrix $N\mathbf{C}^H\mathbf{W}\mathbf{C} - \mathbf{I}$ can be minimized instead [21,22]. Both approaches lead to the same linear system of M equations

$$\mathbf{A}\mathbf{w} = \mathbf{b} \quad (4.52)$$

with

$$\begin{aligned} [\mathbf{A}]_{m,m'} &= \left| [\mathbf{C}\mathbf{C}^H]_{m,m'} \right|^2 \\ [\mathbf{b}]_m &= \frac{1}{N} [\mathbf{C}\mathbf{C}^H]_{m,m}. \end{aligned} \quad (4.53)$$

Due to the often poor condition and the high computational complexity, solving Eq. (4.52) is only practicable for non-Cartesian sampling patterns with small number of samples. It is worth noting that the weights obtained by the least squares approximation of Eq. (4.51) are not necessarily positive anymore.

Similarly,

$$N\mathbf{C}\mathbf{C}^H\mathbf{W} = \mathbf{I} \quad (4.54)$$

is derived by demanding consistency of the signal in k-space rather than of the image in space. If this condition is relaxed by considering the diagonal elements of the resulting matrices on the left and on the right only, it reduces to

$$N\mathbf{C}\mathbf{C}^H\mathbf{w} = \mathbf{1}. \quad (4.55)$$

The fixed-point iteration

$$[\mathbf{w}^{[i+1]}]_m = \frac{[\mathbf{w}^{[i]}]_m}{N[\mathbf{C}\mathbf{C}^H\mathbf{w}^{[i]}]_m} \quad (4.56)$$

was suggested for the solution of Eq. (4.55), which remains practicable for non-Cartesian sampling patterns with higher number of samples using a truncated convolution kernel [23].

A generalization of gridding is obtained by replacing the sampling density compensation with a spatially variant convolution kernel. It promises better accuracy, but requires an optimization of the convolution kernel once per sampling pattern [22].

4.4 Iterative reconstruction

Mainly as a reference for gridding in the later examples, an iterative reconstruction for non-Cartesian imaging based on an NFFT and an adjoint NFFT is derived in this section.

Replacing $\tilde{\mathbf{s}}$ by $\mathbf{s}$ in Eq. (4.31) leads to the forward model

$$\mathbf{s} = \mathbf{E} \mathbf{x}. \quad (4.57)$$

The inverse problem is solved by the least squares approximation

$$\min_{\hat{\mathbf{x}}} \|\mathbf{s} - \mathbf{E} \hat{\mathbf{x}}\|_2^2 \quad (4.58)$$

using the normal equations of the first kind

$$\mathbf{E}^H \mathbf{E} \hat{\mathbf{x}} = \mathbf{E}^H \mathbf{s} \quad (4.59)$$

or by the weighted least squares approximation

$$\min_{\hat{\mathbf{x}}} \sum_{m=1}^M [\mathbf{w}]_m |[s - \mathbf{E} \hat{\mathbf{x}}]_m|^2 \quad (4.60)$$

using the modified normal equations of the first kind

$$\mathbf{E}^H \mathbf{W} \mathbf{E} \hat{\mathbf{x}} = \mathbf{E}^H \mathbf{W} \mathbf{s}. \quad (4.61)$$

Here, Eq. (4.57) is assumed to be overdetermined. Undersampling is considered in subsequent chapters, as well as multiple receive coils. Moreover, the vector $\mathbf{w}$ and the diagonal matrix $\mathbf{W}$ are specifically those introduced in Eq. (4.38) for sampling density compensation.

To determine $\hat{\mathbf{x}}$ in Eq. (4.59) or Eq. (4.61), iterative methods, such as the conjugate gradient (CG) method described in Chapter 3, are preferably employed. They typically involve a repeated calculation of products of the system matrix and a vector $\hat{\mathbf{p}}$. These products can efficiently be computed either with an NFFT and an adjoint NFFT or with a fast convolution [24]. For the latter, the entries of the system matrix are written as

$$[\mathbf{E}^H \mathbf{E}]_{n,n'} = \sum_{m=1}^M e^{i2\pi k_m (r_n - r_{n'})} \quad (4.62)$$

or

$$[\mathbf{E}^H \mathbf{W} \mathbf{E}]_{n,n'} = \sum_{m=1}^M w_m e^{i2\pi k_m (r_n - r_{n'})} \quad (4.63)$$

and referred to as $q(r_n - r_{n'})$. It is worth noting that $q(r)$ closely corresponds to the PSF of a reconstruction with an adjoint NFFT or gridding. The products then reduce to the convolution

$$\sum_{n'=1}^N \hat{p}_{n'} q(r_n - r_{n'}) \quad (4.64)$$

in space, which can be evaluated by a multiplication in k-space. In this way, an NFFT and an adjoint NFFT are essentially replaced by an FFT and an inverse FFT. The Fourier transform of q has to be

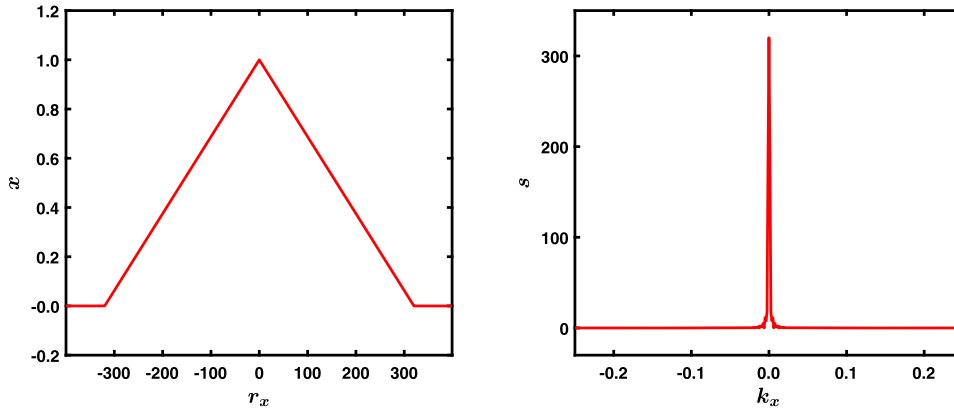


FIGURE 4.5 1D Object

A continuous triangular function, covering 80% of the FOV, is chosen as 1D object (left). Its frequency spectrum is real and nonnegative (right).

computed only once. To prevent backfolding, the fast convolution requires a two-fold oversampling. Nevertheless, it usually provides a substantial acceleration for $M \geq N$ [25].

Iterative methods can synthesize signal outside the k-space area covered by the acquisition [26]. This concerns in particular the corners of a square k-space if only the circle inscribed in the square is actually sampled. Since recovering such signal is ill-conditioned, it typically entails an excessive amplification of high-frequency noise, as discussed in Chapter 2. A low-pass filtering is usually applied to $\hat{\mathbf{x}}$ to avoid this.

4.5 Examples

Gridding is illustrated with a 1D and a 2D example in this section. The triangular function shown in Fig. 4.5 serves as object in the 1D example. The Fourier transform of this object is given by

$$s(k_x) = R \operatorname{sinc}^2(\pi R k_x), \quad (4.65)$$

where $R = 320$.

Initially, an equidistant sampling of s with k_x in the range $[-\frac{1}{4}, \frac{1}{4})$ and $M = 400$ is assumed. It allows applying an inverse FFT for reference and assessing aliasing or truncation errors introduced by gridding. The object is reconstructed at $r_x = -\frac{N}{2}, -\frac{N}{2} + 1, \dots, \frac{N}{2} - 1$ with $N = 800$. Thus, it is enlarged by Fourier interpolation by a factor of 2, involving zero padding in k-space to the range $[-\frac{1}{2}, \frac{1}{2})$. Resulting errors are plotted in Fig. 4.6 and dominated by ringing artifacts at $r_x = -320, 0, 320$. The triangular function is continuous but not differentiable at these points. Therefore, the classical Gibbs phenomenon is not observed, but a poorer approximation by a truncated Fourier series is expected due to the slower decay of the Fourier coefficients with frequency. Compared to the inverse FFT, gridding

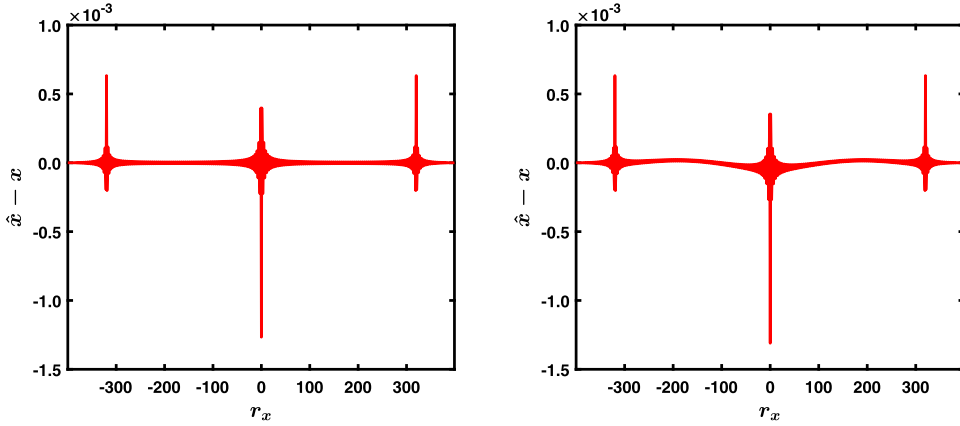


FIGURE 4.6 1D Cartesian Sampling — Inverse FFT versus Gridding

From an equidistant sampling of the frequency spectrum s in Fig. 4.5, 1D objects $\hat{x}$ are reconstructed with twice the spatial resolution using zero padding, once with an inverse FFT (left) and once with gridding (right). Errors are calculated with respect to the continuous triangular function x .

with $f = 2$ and $K = 5$ adds only negligible errors that originate from the truncation of the convolution kernel in this case. In the following, these settings of the oversampling factor and the kernel size are kept.

To simulate a nonequidistant sampling of s , the magnetic field gradient in the frequency-encoding direction, also known as readout gradient, is assumed to ramp up $100\mu\text{s}$ after the excitation with a gradient slew rate of $200\text{Tm}^{-1}\text{s}^{-1}$, at which point in time the data acquisition is also triggered, and to level off at a gradient strength of 40Tm^{-1} , as illustrated in Fig. 4.7. The gradient system is modeled as a linear time-invariant system, once with an ideal frequency response

$$H(i\omega) = 1 \quad (4.66)$$

and once with a more realistic frequency response

$$H(i\omega) = \frac{e^{i\omega T}}{1 + i\omega T}, \quad (4.67)$$

which corresponds to a mono-exponential decay with a time constant T , chosen to be $40\mu\text{s}$ in this example. The amplitude and phase of $H(i\omega)$ are plotted in Fig. 4.8. In the ideal case, the actual gradient waveform is identical to the demand gradient waveform. In the more realistic case, however, the actual gradient waveform is smoothed. This is seen in Fig. 4.7, as well as the resulting k-space sampling positions as a function of time. The data acquisition is performed twice, with positive and negative polarity of the readout gradient, to traverse the 1D k-space from the center, or origin, to the periphery, in positive and negative direction, respectively. Thus, the same range of k_x as before is covered. The sampling positions are located in Fig. 4.9 at the points of intersection of the horizontal lines ($k_y = 0$) with the circles. While the sampling density in principle increases towards the k-space center, $|k_x| < \frac{0.36}{N}$ remains

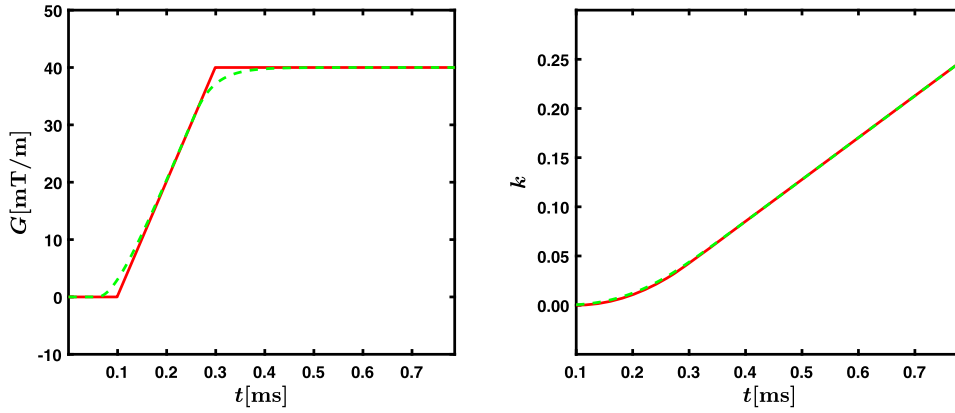


FIGURE 4.7 Gradient Waveform and k-Space Trajectory

The demand gradient waveform (solid red (mid gray in print version), left) linearly increases with the maximum gradient slew rate until the maximum gradient strength is reached. The real gradient system from Fig. 4.8 low-pass filters the gradient waveform (dashed green (gray in print version), left), leading to a distortion of the k-space trajectory (right), which is hardly perceptible at this scale.

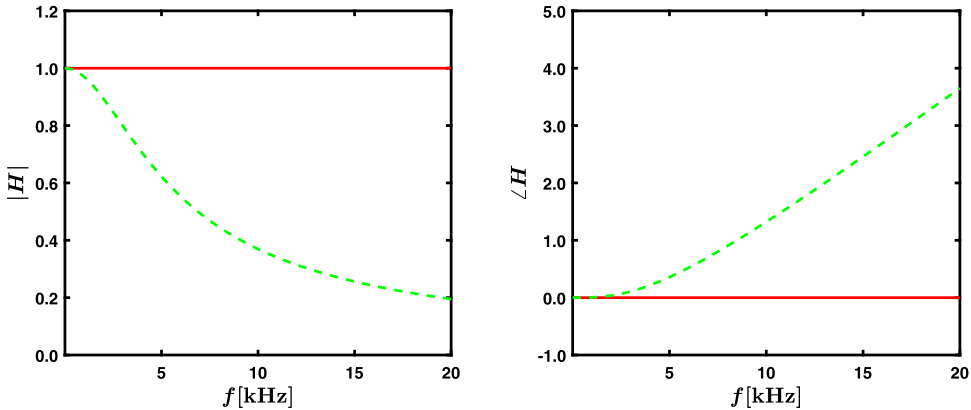


FIGURE 4.8 Gradient System Characterization

An ideal gradient system (solid red (mid gray in print version)) has an invariant amplitude (left) and phase (right) response, whereas a real gradient system (dashed green (gray in print version)) typically exhibits an amplitude response decaying with frequency and a phase response growing with frequency.

uncovered in the more realistic case. It is worth noting that this does not constitute an undersampling per se because the critical sampling distance of $\frac{1}{N}$ is larger than the diameter of the gap.

Applying gridding to the nonequidistant sampling of s produces the errors plotted in Fig. 4.10. With the ideal gradient system, a moderate offset is observed, whereas, with the more realistic gradient system, major errors apparently arise from the gap.

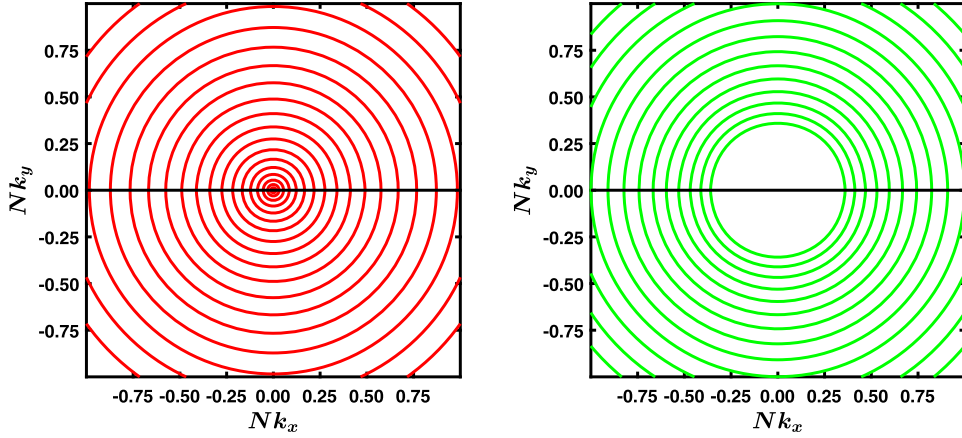


FIGURE 4.9 1D and 2D Sampling Patterns

From the k-space trajectories in Fig. 4.7, 1D (k_x) and 2D (k_x, k_y) sampling patterns are generated by equiangular rotation around the origin. Unlike with the ideal gradient system (left), the sampling pattern obtained with the more realistic gradient system (right) leaves a gap in the k-space center, which is enlarged in both plots.

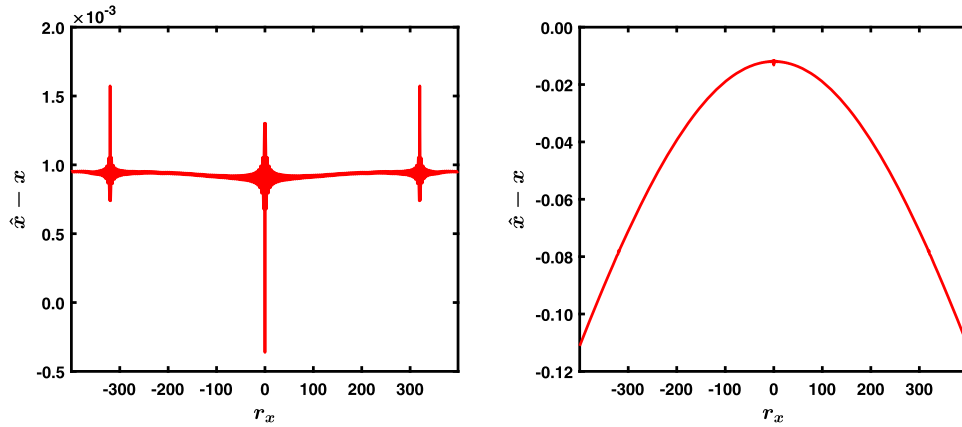


FIGURE 4.10 1D Non-Cartesian Sampling — Gridding

Using the two nonequidistant 1D sampling patterns from Fig. 4.9, 1D objects $\hat{x}$ are reconstructed with twice the spatial resolution with gridding. Errors are calculated with respect to the continuous triangular function x .

So far, a sampling density compensation based on the distance between adjacent k-space sampling positions has been used. Weights obtained with this geometrical approach and with some of the other approaches described in the previous section are juxtaposed in Fig. 4.11, along with corresponding errors, for the problematic case of the more realistic gradient system. The fix-point iteration according to

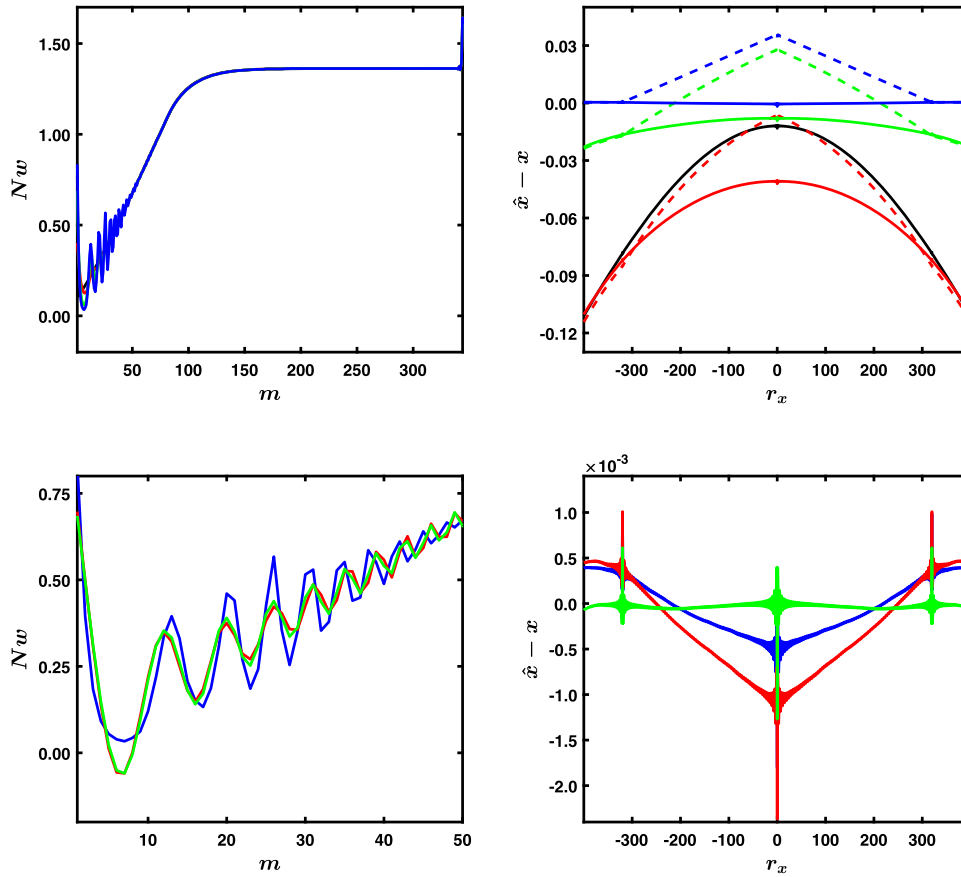


FIGURE 4.11 Sampling Density Compensation

From the second nonequidistant 1D sampling pattern in Fig. 4.9, weights for sampling density compensation are derived (left), geometrically (black, top), numerically based on repeated gridding with 5 (red (mid gray in print version), top), 25 (green (gray in print version), top), and 125 (blue (dark gray in print version), top and bottom) iterations, once with (solid) and once without (dashed) normalization, and algebraically based on Eq. (4.55) (red (mid gray in print version), bottom) and Eq. (4.52) (green (gray in print version), bottom). 1D objects $\hat{x}$ are reconstructed with gridding, and errors are calculated with respect to the continuous triangular function x (right).

Eq. (4.56) slowly converges from $\mathbf{w}^{[0]} = \frac{1}{N} \mathbf{1}$, for which the weights always remain positive. A sinc^2 convolution kernel, truncated to two side lobes, was employed in this example to map nonequidistant samples to nonequidistant samples directly [27]. Errors decrease at first, but soon they are dominated by a systematic overestimation of the weights due to the omitted side lobes. By contrast, enforcing $\|\mathbf{w}\|_1 = 1$ allows reducing errors by two orders of magnitude overall, yet results deteriorate again when proceeding with the iteration. Solving Eq. (4.55) and Eq. (4.52) leads to weights that still show the

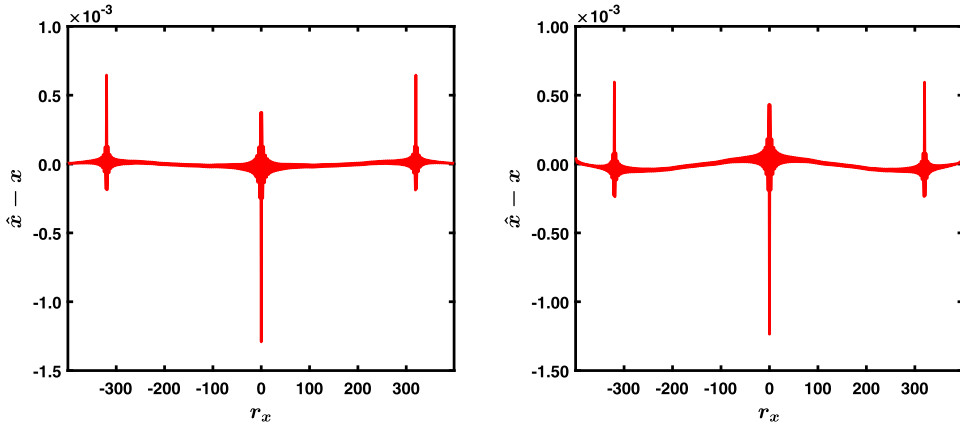


FIGURE 4.12 1D Non-Cartesian Sampling — Iterative Reconstruction

Using the two nonequidistant 1D sampling patterns from Fig. 4.9, 1D objects $\hat{x}$ are iteratively reconstructed with twice the spatial resolution. Errors are calculated with respect to the continuous triangular function x .

striking, unexpected oscillation for the sampling positions in the k -space center, but that are not necessarily positive anymore. The same convolution kernel as before was chosen in the first case, while the Dirichlet kernel from Eq. (4.15) was selected in the second case. A CG method without explicit regularization reliably converges only in the second case, in which also the lowest errors are attained.

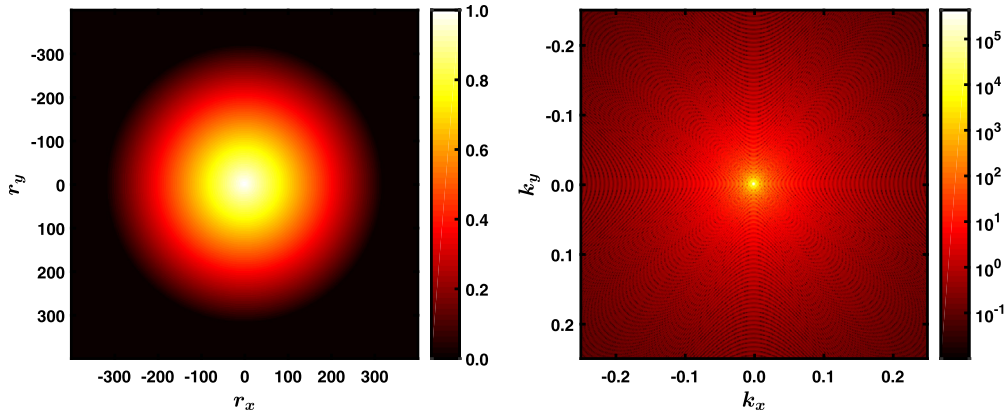
Finally, results obtained with an iterative reconstruction are provided for reference in Fig. 4.12. A CG method, initialized with $\hat{x}^{[0]} = \mathbf{0}$, leads to similar errors as gridding in the case of equidistant sampling and, with favorable sampling density compensation, in the case of nonequidistant sampling. An NFFT and an adjoint NFFT were used in the iterative reconstruction, but no sampling density compensation or explicit regularization.

The conic function shown in Fig. 4.13 is chosen as object to illustrate gridding with a 2D example. Its Fourier transform is given by

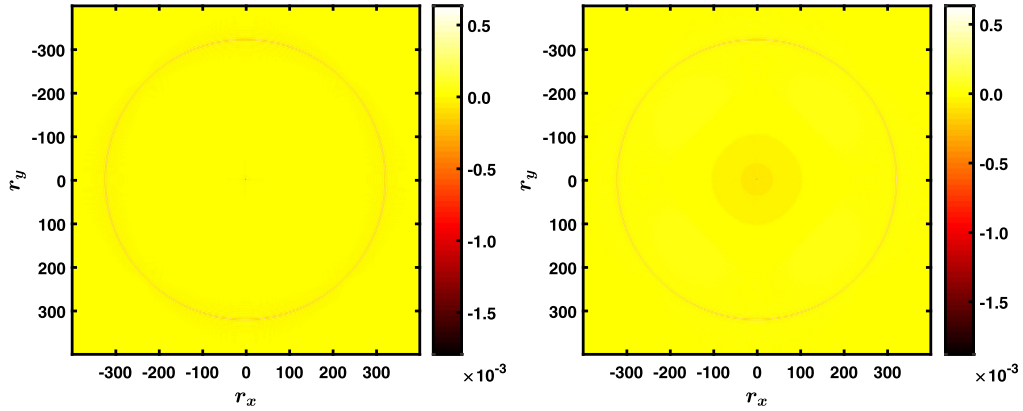
$$s(k_r) = \begin{cases} \pi^2 \frac{J_1(Rk_r)H_0(Rk_r) - J_0(Rk_r)H_1(Rk_r)}{k_r^2}, & k_r \neq 0 \\ \frac{1}{3}\pi R^2, & \text{otherwise} \end{cases}, \quad (4.68)$$

where J_0 , J_1 , and H denote the zeroth and first order Bessel functions of the first kind and the Struve function, respectively [28].

Applying an inverse FFT for reference and gridding to an equidistant sampling of s with k_x and k_y in the range $[-\frac{1}{4}, \frac{1}{4}]$ and $M = 400^2$ results in the errors plotted in Fig. 4.14, which are dominated by ringing artifacts at $r_x^2 + r_y^2 = 320^2$ and at the origin, where the conic function is not differentiable. Additional errors introduced by gridding remain negligible, even though they are perceptibly higher than in the 1D example. Gridding requires about 2.75 times the number of floating point operations of the inverse FFT in this case, of which more than 80% are involved in the Fourier transform due to the oversampling.

**FIGURE 4.13 2D Object**

A continuous conic function, covering 50% of the square FOV and 64% of the inscribed circular FOV, serves as 2D object (left). The magnitude of its frequency spectrum is displayed on a logarithmic scale (right).

**FIGURE 4.14 2D Cartesian Sampling — Inverse FFT versus Gridding**

From an equidistant sampling of the frequency spectrum in Fig. 4.13, 2D objects are reconstructed with twice the spatial resolution using zero padding, once with an inverse FFT (left) and once with gridding (right). Errors are calculated with respect to the continuous conic function.

The same readout gradient as in the 1D example is assumed to simulate a nonequidistant sampling of s , but the data acquisition is repeated $\lceil \pi \sqrt{M} \rceil = 1257$ times. The direction of the readout gradient is rotated about the k -space origin by a constant angle after each instance, and with it the sampling positions in the 2D k -space. The number of instances is selected such that the distance between adjacent sampling positions in the azimuthal direction does not exceed $\frac{1}{2\sqrt{M}}$. Thus, undersampling is avoided. The resulting sampling patterns, which cover a circular k -space area, are shown in Fig. 4.9, where the

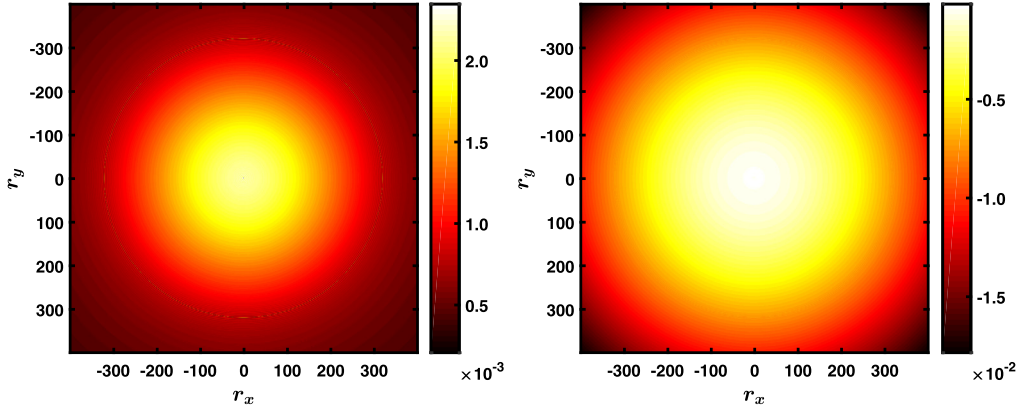


FIGURE 4.15 2D Non-Cartesian Sampling — Gridding

Using the two nonequidistant 2D sampling patterns from Fig. 4.9, 2D objects are reconstructed with twice the spatial resolution with gridding. Errors are calculated with respect to the continuous conic function.

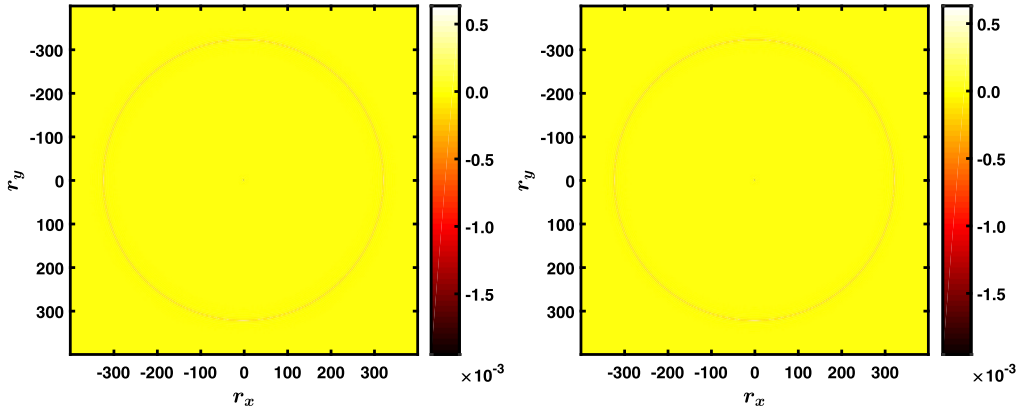


FIGURE 4.16 2D Non-Cartesian Sampling — Iterative Reconstruction

From the two nonequidistant 2D sampling patterns in Fig. 4.9, 2D objects are iteratively reconstructed with twice the spatial resolution. Errors are calculated with respect to the continuous conic function.

seemingly continuous sampling in the azimuthal direction is due to the high sampling density in the k -space center.

Applying gridding and an iterative reconstruction to the nonequidistant sampling of s produces the errors plotted in Fig. 4.15 and Fig. 4.16, respectively. Weights for sampling density compensation were derived with a geometrical approach and only employed in gridding. Most notably, the errors obtained with gridding in the case of the more realistic gradient system are reduced by one order of magnitude relative to the 1D example.

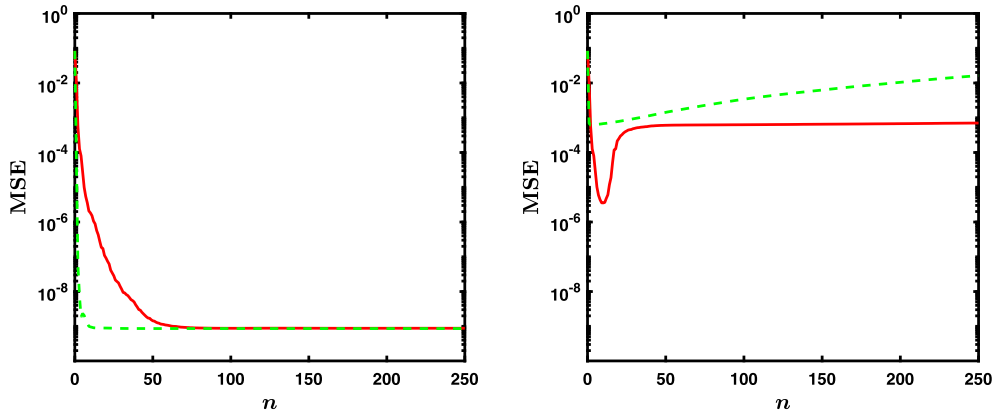


FIGURE 4.17 2D Non-Cartesian Sampling — Convergence

2D objects are iteratively reconstructed without (solid red (mid gray in print version)) and with (dashed green (gray in print version)) sampling density compensation, once without (left) and once with (right) added noise emulating a nominal SNR of 50. Mean squared errors with respect to the continuous conic function are plotted as function of the number of iterations.

The convergence of the iterative reconstruction is analyzed in Fig. 4.17. While using Eq. (4.61) with a sampling density compensation instead of Eq. (4.59) without a sampling density compensation leads to considerable acceleration, it adversely affects the convergence in the presence of noise, which demands either a suitable explicit regularization or a sufficiently early termination of the iteration.

4.6 Spatial resolution and noise

Especially when comparing non-Cartesian imaging with Cartesian imaging, awareness of potential differences in spatial resolution and noise is essential and, therefore, raised in this section.

A common measure of spatial resolution is the full width at half maximum (FWHM) of the main lobe of the PSF. In 1D Cartesian imaging, the PSF is given by Eq. (4.37) and approximated by

$$\text{PSF}(r) \approx \text{sinc}(\pi r) \quad (4.69)$$

for $M = N$ and small $\frac{r}{N}$. The FWHM equals 1.21 since $\text{sinc}(0.603\pi) = 0.5$. In 2D Cartesian imaging, the PSF remains the same along the axes r_x and r_y , but changes to

$$\text{PSF}(r_d) \approx \text{sinc}^2\left(\frac{\pi}{\sqrt{2}}r_d\right) \quad (4.70)$$

along the diagonals r_d , with $r_d = \sqrt{2}r_x$ and $r_y = \pm r_x$, assuming a square k-space area to be covered by the acquisition. The FWHM increases to 1.25, which indicates a slightly anisotropic spatial resolution.

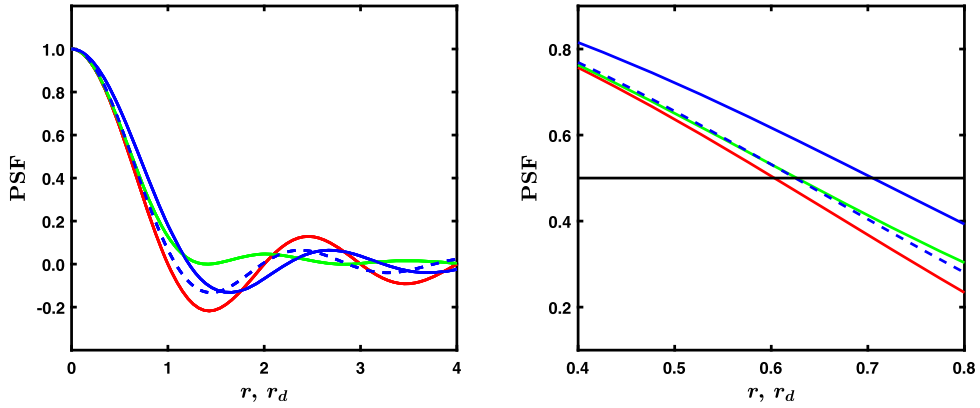


FIGURE 4.18 Spatial Resolution

PSFs are approximated for acquisitions covering a square FOV, along the axes (solid red (mid gray in print version)) and the diagonals (solid green (gray in print version)), and circular FOVs, with the circle inscribed in the square (solid blue (dark gray in print version)) and the area of the circle and the square equalized (dashed blue (dark gray in print version)). Crossings of the main lobes with the horizontal line at half the maximum amplitude are determined.

In 2D non-Cartesian imaging, only a smaller circular k-space area is usually covered, as seen in Fig. 4.1. The PSF is approximated by

$$\text{PSF}(r) \approx 2 \text{jinc}(\pi r) \quad (4.71)$$

using

$$\text{jinc}(x) = \begin{cases} \frac{J_1(x)}{x}, & x \neq 0 \\ \frac{1}{2}, & \text{otherwise} \end{cases}, \quad (4.72)$$

and the FWHM amounts to 1.41 regardless of direction. Eq. (4.71), as Eq. (4.69) and Eq. (4.70), can be derived by assuming a continuous sampling of the covered k-space area [29].

To match the spatial resolution in 2D Cartesian imaging, the circular k-space area has to be enlarged relative to the square k-space area. Scaling the radius of the former by $\frac{2}{\sqrt{\pi}} = 1.13$ equalizes the area of both [30]. Applied to Eq. (4.71), it results in

$$\text{PSF}(r) \approx 2 \text{jinc}(2\sqrt{\pi} r) \quad (4.73)$$

and an FWHM of 1.25. This is illustrated in Fig. 4.18, in which the PSFs according to Eq. (4.69), Eq. (4.70), Eq. (4.71), and Eq. (4.73) are plotted. It is worth noting that the amplitudes of the first side lobes of the PSF along the axes in 2D Cartesian imaging are the highest. In general, this leads to more pronounced ringing artifacts in 2D Cartesian imaging than in 2D non-Cartesian imaging.

The validity of the continuous sampling approximation is substantiated in Fig. 4.19. Despite the rather small number of samples, the calculated PSFs for PROPELLER, radial, and spiral imaging using

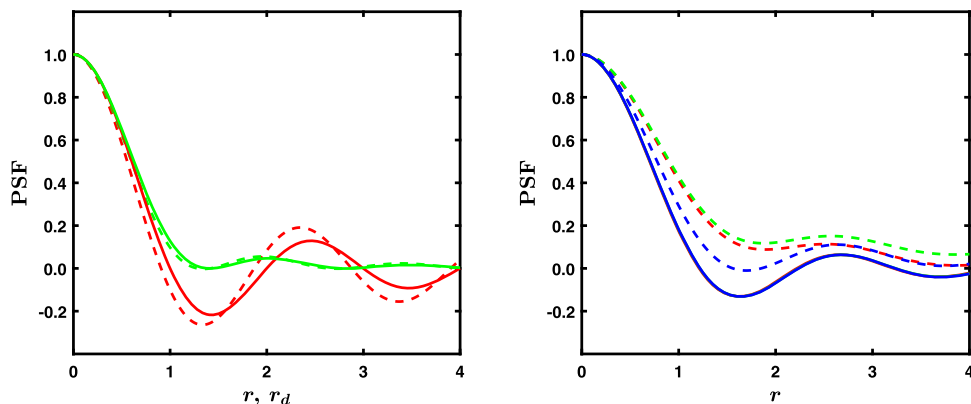


FIGURE 4.19 Point Spread Functions

PSFs are calculated for the sampling patterns in Fig. 4.1, namely for EPI along the axes (red (mid gray in print version), left) and the diagonals (green (gray in print version), left), as well as for PROPELLER (red (mid gray in print version), right), radial (green (gray in print version), right), and spiral (blue (dark gray in print version), right) imaging, and gridding without (dashed) and with (solid) sampling density compensation.

gridding with sampling density compensation closely agree with each other, to the extent that they are not discernible at this scale, and with the approximated PSF in Fig. 4.18. The calculated and approximated PSFs for EPI along the axes and the diagonals are similarly consistent. Naturally, this only holds for the main lobe and the first side lobes.

While the sampling density compensation is crucial to shape the PSF in non-Cartesian imaging using gridding, it entails a reduction in SNR by a factor of

$$\frac{\sum_{m=1}^M [\mathbf{w}]_m}{\sqrt{M} \|\mathbf{w}\|_2}, \quad (4.74)$$

assuming additive white Gaussian noise [31]. This factor equals 1.0 only if the weights are uniform. For the 2D sampling patterns in Fig. 4.1, it decreases to 0.98, 0.85, 0.87, and 0.96 in the case of echo-planar, PROPELLER, radial, and spiral imaging, respectively. It drops further from 0.87 to 0.80 for the 2D sampling patterns in Fig. 4.9 due to the additional nonequidistant sampling in radial direction.

The same reduction in SNR is observed in the right plot in Fig. 4.17 using an iterative reconstruction without sampling density compensation. The nominal SNR of 50 accounts for a mean squared error of $4 \cdot 10^{-4}$, given that the maximum signal is normalized to a value of 1.0 and that other errors are negligible. The actual mean squared error of $6.3 \cdot 10^{-4}$ after 100 iterations then implies an amplification of the noise variance by a factor of 1.58 and a loss in SNR of 20%. However, a CG method provides a diminishing intrinsic regularization in the course of the iterations [32], which is particularly evident using an iterative reconstruction with sampling density compensation. The amplification of the noise variance is ultimately given by the diagonal elements of the matrix

$$\mathbf{X} = (\mathbf{E}^H \mathbf{E})^{-1} \quad (4.75)$$

or

$$\mathbf{X} = (\mathbf{E}^H \mathbf{W} \mathbf{E})^{-1} (\mathbf{E}^H \mathbf{W}^2 \mathbf{E}) (\mathbf{E}^H \mathbf{W} \mathbf{E})^{-1} \quad (4.76)$$

and is often unacceptably high without explicit regularization because the system matrix is typically ill-conditioned in non-Cartesian imaging [26].

Variation in sampling density also leads to colored noise, which is characterized by a nonconstant power spectral density. Colored noise notably impairs the perceived spatial resolution of images [33]. This can be mitigated by adding noise or by considering other criteria, for instance contrast, in the sampling density compensation [34].

4.7 Extensions

An NFFT for an efficient conversion of equidistant samples in space into nonequidistant samples in k -space was introduced in a previous section. Moreover, an adjoint NFFT was shown to be an integral part of gridding and, together with an NFFT, of an iterative reconstruction for non-Cartesian imaging. Other NFFT exist for related conversions, which are of relevance to both Cartesian and non-Cartesian imaging. They are outlined in this section, as well as their applications.

For a mapping of nonequidistant samples in space to equidistant samples in k -space, the approximation

$$e^{-i2\pi kr} \approx \frac{1}{f \tilde{c}_{-Nk}} \sum_{n'=-fN/2}^{fN/2-1} \tilde{c}\left(r - \frac{n'}{f}\right) e^{-i2\pi k \frac{n'}{f}} \quad (4.77)$$

is obtained similarly to Eq. (4.11), taking the different range of r and k and the necessary stretching and scaling of $\tilde{c}$ into account. It leads to the forward model

$$\tilde{s}_n \approx \frac{1}{f \tilde{c}_{-Nk_n}} \sum_{n'=-fN/2}^{fN/2-1} \left(\sum_{m=1}^M x_m \tilde{c}\left(r_m - \frac{n'}{f}\right) \right) e^{-i2\pi k_n \frac{n'}{f}}, \quad (4.78)$$

where $k_n = -\frac{1}{2}, -\frac{1}{2} + \frac{1}{N}, \dots, \frac{1}{2} - \frac{1}{N}$. The inner sum converts M nonequidistant samples in space by a convolution into fN equidistant samples in space. The outer sum amounts to an FFT of length fN , transforming the equidistant samples in space into equidistant samples in k -space. The deapodization is applied last.

Nonequidistant samples in space occur in the presence of an imperfect, namely a spatially nonconstant, magnetic field gradient, which affects the spatial encoding and leads to distortions in images if ignored in the reconstruction. The signal measured in k -space is modeled under these circumstances by

$$s(k) = \int_{-N/2}^{N/2} x(r) e^{-i2\pi kg(r)} dr \quad (4.79)$$

with a usually nonlinear function g mapping the true position r to the apparent position r' in space. Provided that g is invertible, a change in variables results in

$$s(k) = \int_{g(-N/2)}^{g(N/2)} x(g^{-1}(r')) \frac{dg^{-1}(r')}{dr'} e^{-i2\pi kr'} dr' \quad (4.80)$$

and a spatial discretization in nonequidistant samples of x . Currently, a distortion correction is routinely performed after the reconstruction in space, using interpolation and scaling. However, integrating it into the reconstruction promises advantages, particularly with regard to spatial resolution [35]. While this demands an efficient evaluation of Eq. (4.78) in Cartesian imaging, the forward model has to be generalized to nonequidistant samples in both domains in non-Cartesian imaging.

Nonequidistant samples in space also arise from an imperfect main magnetic field. The signal is described by

$$s(t) = \int_{-N/2}^{N/2} x(r) e^{-i2\pi k(t)r} e^{-i2\pi t f(r)} dr \quad (4.81)$$

in this case. Here, s and k are explicit functions of t , the time after the excitation, and f represents the offset in the frequency of the precession of the nuclear spins induced by the spatial variation of the main magnetic field. In Cartesian imaging, k is a linear function because the readout gradient remains unchanged during the data acquisition. Eq. (4.81) can then be written as Eq. (4.79). In EPI, this still holds in the phase-encoding direction, in which the dominant distortions occur [36].

The consideration of an inhomogeneous main magnetic field in non-Cartesian imaging is treated in Chapter 12. Only an NFFT for an efficient conversion between nonequidistant samples in both domains is derived here, which was shown to be applicable to such an off-resonance correction [37,38]. The inverse Fourier transform of the window function c

$$(\mathcal{F}^{-1}c)(r) = \int_{-\infty}^{\infty} c(k) e^{i2\pi kr} dk \quad (4.82)$$

is rewritten as

$$(\mathcal{F}^{-1}c)(r) = \int_{-1/2}^{1/2} \sum_{p=-\infty}^{\infty} c(k+p) e^{i2\pi(k+p)r} dk \quad (4.83)$$

and approximated by

$$(\mathcal{F}^{-1}c)(r) \approx \frac{1}{fN} \sum_{n'=-fN/2}^{fN/2-1} \sum_{p=-\infty}^{\infty} c\left(k - \frac{n'}{fN} + p\right) e^{i2\pi(k - \frac{n'}{fN} + p)r} \quad (4.84)$$

to obtain

$$e^{-i2\pi kr} \approx \frac{1}{fN(\mathcal{F}^{-1}c)(r)} \sum_{n'=-fN/2}^{fN/2-1} \sum_{p=-\infty}^{\infty} c\left(k - \frac{n'}{fN} + p\right) e^{-i2\pi(\frac{n'}{fN} - p)r}. \quad (4.85)$$

The inner sum is then eliminated, for example by simply limiting the range of c to $[-\frac{K}{2fN}, \frac{K}{2fN}]$ and of k to $[-\frac{1}{2} + \frac{K}{2fN}, \frac{1}{2} - \frac{K}{2fN}]$, leading to

$$e^{-i2\pi kr} \approx \frac{1}{fN(\mathcal{F}^{-1}c)(r)} \sum_{n'=-fN/2}^{fN/2-1} c'\left(k - \frac{n'}{fN}\right) e^{-i2\pi \frac{n'}{fN}r}. \quad (4.86)$$

Such an NFFT also permits including higher order magnetic field perturbations in the reconstruction, which can be modeled by

$$s(t) = \int_{-N/2}^{N/2} x(r) e^{-i2\pi \sum_{l=1}^L k_l(t)b_l(r)} dr \quad (4.87)$$

with spatial basis functions b and corresponding time-variant coefficients k [39]. While spherical harmonics are commonly chosen for b , separate calibration measurements, such as gradient impulse response measurements, or concurrent magnetic field monitoring are usually required to determine k [40,41]. The first-order spherical harmonics are equivalent to spatially constant magnetic field gradients and thus describe the actual sampling positions in k -space. Accurate knowledge of the corresponding time-variant coefficients is often necessary but also sufficient to achieve adequate image quality in non-Cartesian imaging.

Finally, exponential signal decay during the data acquisition can be incorporated into Eq. (4.81) by adding a suitably scaled decay rate as imaginary part to f [42].

4.8 Summary

NFFTs are an indispensable part of non-Cartesian MRI reconstruction. They provide a generalization of FFTs to nonequidistant samples and rely on an approximation with controllable accuracy, which is, even with small oversampling factors and kernel sizes, sufficient for MRI. Several optimized implementations are nowadays available for various computing platforms, also under other acronyms, such as NUFFTs and FINUFFTs. They considerably facilitate the development and application of general algorithms coping with various sampling patterns and further alleviate concerns about the running time of NFFTs.

Two such algorithms were described in this chapter, gridding and an iterative reconstruction. Gridding involves, in addition to an adjoint NFFT, a sampling density compensation, with which reasonable accuracy is attained for most popular sampling patterns. The iterative reconstruction comprises both an NFFT and an adjoint NFFT and allows achieving better accuracy at the expense of computational complexity. Both algorithms form the basis for other algorithms discussed in following chapters, which notably support an acceleration of non-Cartesian imaging by undersampling.

NFFTs were also demonstrated to be a valuable part of corrections for an imperfect main magnetic field and magnetic field gradient. Rendering these corrections more robust, in particular without accurate knowledge of nuisance parameters, still constitutes a considerable challenge.

While many non-Cartesian imaging methods have been proposed and investigated to date, only a few have been adopted in clinical routine on a larger scale so far. Whether the use of non-Cartesian

imaging methods will further expand and ever rival the use of Cartesian imaging methods remains to be seen.

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